

Privacy Shield

Chrome Extension for Privacy & Ad Blocking

Introduction

Privacy Shield is a Manifest V3 compliant Chrome browser extension designed to empower users with granular control over their browsing privacy. The extension leverages Chrome's Declarative Net Request API to block known tracking domains, advertisements, and analytics scripts in real-time. This project emerged from the necessity of creating a lightweight, performant privacy solution without sacrificing functionality. Privacy Shield enables users to manage blocklists, whitelists, and custom rules through an intuitive popup interface and comprehensive settings panel, providing transparent insight into blocked requests via a badge counter.

Abstract

Privacy Shield delivers real-time tracking prevention and ad blocking through declarative net request rules. The extension manages 20 core blocking rules targeting major ad networks and analytics platforms including Google Analytics, DoubleClick, Facebook Pixels, and Amazon Ad System. Users can dynamically create allow and block rules, customize domain filtering, and monitor blocking statistics. The solution consists of 488 lines of JavaScript across four modules: background service worker (80 LOC), popup interface (141 LOC), options page (267 LOC), plus static rule configuration. The extension operates on Manifest V3 standards and maintains local storage for user preferences, custom rules, and statistics. Version 1.0.1 includes visual feedback mechanisms, rule persistence, and a clean dark-themed UI for enhanced usability.

Tools Used

Tool	Purpose
JavaScript (ES6+)	Core logic for background worker, popup, options pages
Chrome API (DNR)	Declarative Net Request for rule enforcement
Chrome Storage API	Persistent user preferences and statistics
HTML5 / CSS3	UI markup and styling for popup and options
Manifest V3	Extension configuration and permissions
JSON	Rule definitions and configuration storage

Steps Involved in Building the Project

- 1. Planning & Architecture:** Designed extension architecture around Manifest V3 standards with focus on performance and user control. Identified core requirement of declarative net request implementation over deprecated content script blocking.
- 2. Rule Definition:** Curated 20 primary blocking rules targeting high-impact tracker domains including Google Analytics, Facebook Pixels, DoubleClick, Criteo, Taboola, Hotjar, Mixpanel, and advertising networks. Configured resource type filters to optimize blocking scope.
- 3. Background Service Worker Development:** Implemented background.js to manage dynamic rule updates, handle rule lifecycle operations, and maintain request count statistics. Utilized Chrome's runtime messaging for inter-component communication.
- 4. Popup UI Implementation:** Created popup.html and popup.js to display real-time blocking counter, current tab statistics, rule management interface, and extension toggle. Implemented dynamic rule rendering with add/remove functionality for user-defined rules.
- 5. Options Page Development:** Built comprehensive options.html and options.js enabling users to customize domain lists, manage allowlist and blocklist, and view aggregate statistics. Implemented persistent storage and user preference management.
- 6. Storage & State Management:** Configured Chrome Storage API for non-volatile persistence of user settings, custom rules, and blocking statistics across browser sessions.
- 7. Testing & Refinement:** Validated rule application, tested dynamic rule creation and removal, verified cross-tab state consistency, and optimized badge counter display logic. Resolved edge cases in rule filtering and storage operations.

Conclusion

Privacy Shield demonstrates practical application of modern browser extension development patterns within Manifest V3 constraints. The project successfully combines lightweight architecture with powerful privacy controls, allowing users to maintain transparency and agency over their browsing data. By implementing declarative net request based blocking instead of content scripts, Privacy Shield achieves superior performance with minimal resource footprint. The extension architecture enables extensibility for future enhancements including advanced rule patterns, domain-specific policies, and privacy analytics. This project serves as a solid foundation for portfolio demonstration of full-stack extension development, JavaScript competency, and understanding of web privacy mechanisms. Deployment to Chrome Web Store would require additional security review and store submission processes.